```
//Tune notes

#define NOTE_C4 262
#define NOTE_D4 294
#define NOTE_E4 330
#define NOTE_F4 349
#define NOTE_G4 392
#define NOTE_A4 440
#define NOTE_C5 523
#define REDLED 10
#define GREENLED 3
#define BLUELED 13
#define YELLOWLED 5
#define REDBUTTON 9
#define GREENBUTTON 2
#define BLUEBUTTON 12
#define YELLOWBUTTON 6
#define BUZZER1 4
#define BUZZER2 7
#define NONE 0
#define RED (1 << 0)
#define GREEN (1 << 1)
#define BLUE (1 << 2)
#define YELLOW (1 << 3)
#define MAXROUNDS 13
#define TIMELIMIT 3000
byte gameBoard[32];
byte gameround = 0;
int melody[] = {NOTE_C4, NOTE_E4, NOTE_G4, NOTE_C5, NOTE_G4, NOTE_E4, NOTE_D4,
0};
int noteDuration = 120;
void setup() {
pinMode(REDBUTTON, INPUT_PULLUP);
pinMode(GREENBUTTON, INPUT_PULLUP);
pinMode(BLUEBUTTON, INPUT_PULLUP);
pinMode(YELLOWBUTTON, INPUT_PULLUP);
pinMode(REDLED, OUTPUT);
pinMode(GREENLED, OUTPUT);
pinMode(BLUELED, OUTPUT);
pinMode(YELLOWLED, OUTPUT);
pinMode(BUZZER1, OUTPUT);
pinMode(BUZZER2, OUTPUT);
digitalWrite(BUZZER1, LOW);
digitalWrite(BUZZER2, LOW);
```

```
play_starting_tune();
wait_for_any_button();
}
void loop() {
attractMode();
showled(RED | GREEN | BLUE | YELLOW);
delay(800);
showled(NONE);
delay(200);
if (play_memory()) win_effect();
else lose_effect();
}


bool play_memory() {
randomSeed(millis());
gameround = 0;
while (gameround < MAXROUNDS) {
add_move();
play_moves();
for (byte i = 0; i < gameround; i++) {
byte input = wait_button();
if (input == NONE) return false;
if (input != gameBoard[i]) return false;
}
delay(400);
}
return true;
}
void add_move() {
byte r = random(0, 4);
if (r == 0) gameBoard[gameround++] = RED;
else if (r == 1) gameBoard[gameround++] = GREEN;
else if (r == 2) gameBoard[gameround++] = BLUE;
else gameBoard[gameround++] = YELLOW;
}
void play_moves() {
for (byte i = 0; i < gameround; i++) {
toner(gameBoard[i], 150);
delay(100);
}
}
byte wait_button() {
long start = millis();
```

```
while (millis() - start < TIMELIMIT) {
byte b = readbutton();
if (b != NONE) {
toner(b, 120);
while (checkButton() != NONE) {
delay(5);
}
delay(10);
return b;
}
}
return NONE;
}
byte readbutton() {
if (!digitalRead(REDBUTTON)) return RED;
if (!digitalRead(GREENBUTTON)) return GREEN;
if (!digitalRead(BLUEBUTTON)) return BLUE;
if (!digitalRead(YELLOWBUTTON)) return YELLOW;
return NONE;
}
byte checkButton() {
return readbutton();
}
void showled(byte v) {
digitalWrite(REDLED, v & RED);
digitalWrite(GREENLED, v & GREEN);
digitalWrite(BLUELED, v & BLUE);
digitalWrite(YELLOWLED, v & YELLOW);
}
void toner(byte which, int buzz_length_ms) {
showled(which);
switch (which) {
case RED: buzz_sound(buzz_length_ms, 1136); break;
case GREEN: buzz_sound(buzz_length_ms, 568); break;
case BLUE: buzz_sound(buzz_length_ms, 851); break;
case YELLOW: buzz_sound(buzz_length_ms, 638); break;
}
showled(NONE);
digitalWrite(BUZZER1, LOW);
digitalWrite(BUZZER2, LOW);
}
void buzz_sound(int buzz_length_ms, int buzz_delay_us) {
long buzz_length_us = buzz_length_ms * 1000L;
while (buzz_length_us > (buzz_delay_us * 2)) {
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```
buzz_length_us -= buzz_delay_us * 2;
digitalWrite(BUZZER1, LOW);
digitalWrite(BUZZER2, HIGH);
delayMicroseconds(buzz_delay_us);
digitalWrite(BUZZER1, LOW);
digitalWrite(BUZZER2, LOW);
delayMicroseconds(buzz_delay_us);
}
digitalWrite(BUZZER1, LOW);
digitalWrite(BUZZER2, LOW);
}
void win_effect() {
for (int i = 0; i < 3; i++) {
showled(GREEN | BLUE);
buzz_sound(150, 400);
showled(RED | YELLOW);
buzz_sound(150, 400);
}
showled(NONE);
}
void lose_effect() {
showled(RED | GREEN);
buzz_sound(250, 1500);
showled(BLUE | YELLOW);
buzz_sound(250, 1500);
showled(NONE);
}
void attractMode() {
while (1) {
showled(RED); delay(80); if (readbutton()) return;
showled(BLUE); delay(80); if (readbutton()) return;
showled(GREEN); delay(80); if (readbutton()) return;
showled(YELLOW); delay(80); if (readbutton()) return;
}
}
void play_starting_tune() {
while (readbutton() == NONE) {
for (int i = 0; i < 8; i++) {
if (melody[i] == NOTE_C4) showled(RED);
else if (melody[i] == NOTE_E4) showled(GREEN);
else if (melody[i] == NOTE_G4) showled(BLUE);
else if (melody[i] == NOTE_C5) showled(YELLOW);
else showled(NONE);
tone(BUZZER2, melody[i], noteDuration);
```

```
delay(noteDuration * 1.3);
noTone(BUZZER2);
showled(NONE);
}
}
showled(NONE);
}
void wait_for_any_button() {
showled(RED | GREEN | BLUE | YELLOW);
while (readbutton() == NONE);
while (readbutton() != NONE);
showled(NONE);
}
```